

GNU nano 7.2

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#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytesRead, bytesWritten;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    src = open(argv[1], O_RDONLY);

    if (src < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest < 0)
    {
        perror("Error opening destination file");
        close(src);
        return 1;
    }

    while ((bytesRead = read(src, buffer, BUFFER_SIZE)) > 0)
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while ((bytesRead = read(src, buffer, BUFFER_SIZE)) > 0)
{
    bytesWritten = write(dest, buffer, bytesRead);

    if (bytesWritten != bytesRead)
    {
        perror("Error writing file");
        close(src);
        close(dest);
        return 1;
    }
}

if (bytesRead < 0)
{
    perror("Error reading file");
}

close(src);
close(dest);

printf("File copied successfully.\n");

return 0;
}|

```

```

sravya@Sravya1955:~/os$ time ./filecopy2 source.txt destination2.txt
File copied successfully.

real    0m0.009s
user    0m0.001s
sys     0m0.008s
sravya@Sravya1955:~/os$ nano filecopy_std.c
sravya@Sravya1955:~/os$ |

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sravya@Sravya1955:~/os$ cat filecopy_std.c
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    FILE *src, *dest;
    char buffer[BUFFER_SIZE];
    size_t bytesRead;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    src = fopen(argv[1], "rb");

    if (src == NULL)
    {
        perror("Error opening source file");
        return 1;
    }

    dest = fopen(argv[2], "wb");

    if (dest == NULL)
    {
        perror("Error opening destination file");
        fclose(src);
        return 1;
    }

    while ((bytesRead = fread(buffer, 1, BUFFER_SIZE, src)) > 0)
    {
        fwrite(buffer, 1, bytesRead, dest);
    }
}
```

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    if (dest == NULL)
    {
        perror("Error opening destination file");
        fclose(src);
        return 1;
    }

    while ((bytesRead = fread(buffer, 1, BUFFER_SIZE, src)) > 0)
    {
        fwrite(buffer, 1, bytesRead, dest);
    }

    fclose(src);
    fclose(dest);

    printf("File copied successfully.\n");

    return 0;
}
sravya@Sravya1955:~/os$ |

```

```

sys      0m0.008s
sravya@Sravya1955:~/os$ nano filecopy_std.c
sravya@Sravya1955:~/os$ gcc filecopy_std.c -o filecopy_std
sravya@Sravya1955:~/os$ ./filecopy_std source.txt destination_std.txt
File copied successfully.
sravya@Sravya1955:~/os$ cat destination_std.txt
Hello!
This is Operating Systems Lab.
Learning system calls in Linux.
sravya@Sravya1955:~/os$ time ./filecopy2 source.txt test1.txt
File copied successfully.

real    0m0.096s
user    0m0.000s
sys     0m0.031s
sravya@Sravya1955:~/os$ time ./filecopy_std source.txt test2.txt
File copied successfully.

real    0m0.020s
user    0m0.001s
sys     0m0.004s
sravya@Sravya1955:~/os$ cat filecopy_std.c

```